Dagens tekst

- Kravspecifikationer
  - Funktionelle og non-funktionelle krav
- Kvalitetssikting
  - Kriterier/måling
  - Placering i udviklingsmodellen
  - Teknikker
  - Statisk og dynamisk kvalitetssikring
  - - i projekplanen
- Klynge 2
- Midtvejsevaluering

Kravspecifikation

Krav og kravspecifikationer

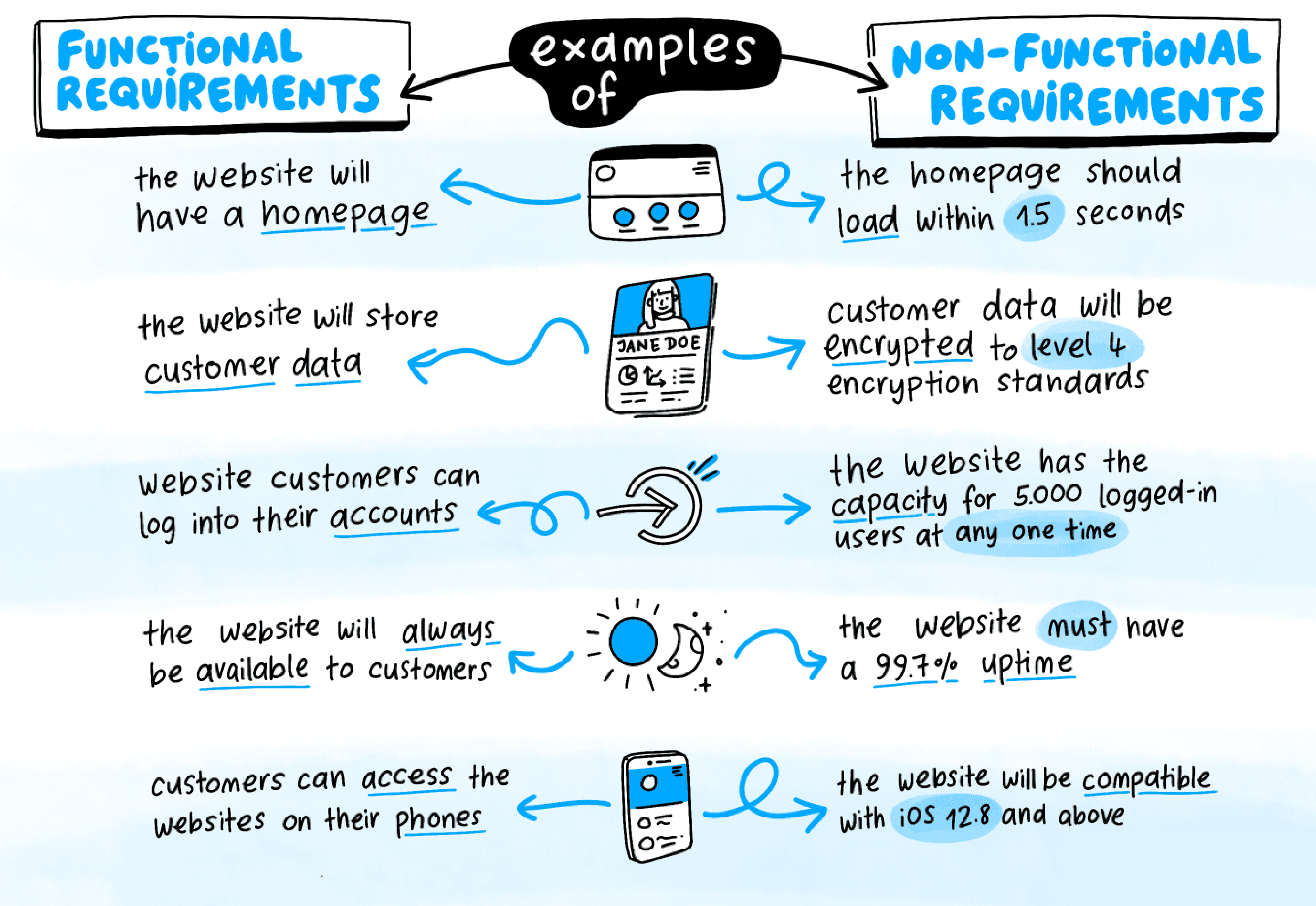
Kravspecifikation. ”everything that the set of relevant stakeholders want from a system” (Avison & Fitzgerald, *Information Systems Development*, 4th ed. McGraw-Hill, 2006, p. 97)

Functional requirements. These are statements of service the system should provide, how the system should react to particular inputs and how the system should behave in particular situations. In some cases the functional requirements may also explicitly state what the system should not do.

Non-functional requirements. These are constraints on the services or functions offered by the system. They include timing constraints, constraints on the development process, standards, etc.

Domain requirements. These are requirements that come from the application domain of the system and that reflect characteristics of that domain. They may be functional or non-functional requirements.

(Sommerville,I. *Software Engineering,* 6th ed., Addison-Wesley, 2000)



https://plan.io/blog/non-functional-requirements-nfrs/

User Story

- A commonly used technique for expressing information systems requirement ideas ("a promise of conversation")
- Three parts of a user story:
  - 1) a specific user role (or **persona**); 2) the user's goal in the user role; and 3) a justification for the goal
- Recipe:
  - *As a <name of the user role>, I want to <the user's goal in the user role> so that <a justification for the goal.>*
- Example of a user story at a high level of abstraction
  - *As a customer within a retail store, I want to be able to easily find a product that fits my needs so that I can complete picking up my purchases accurately, efficiently and without delays.*

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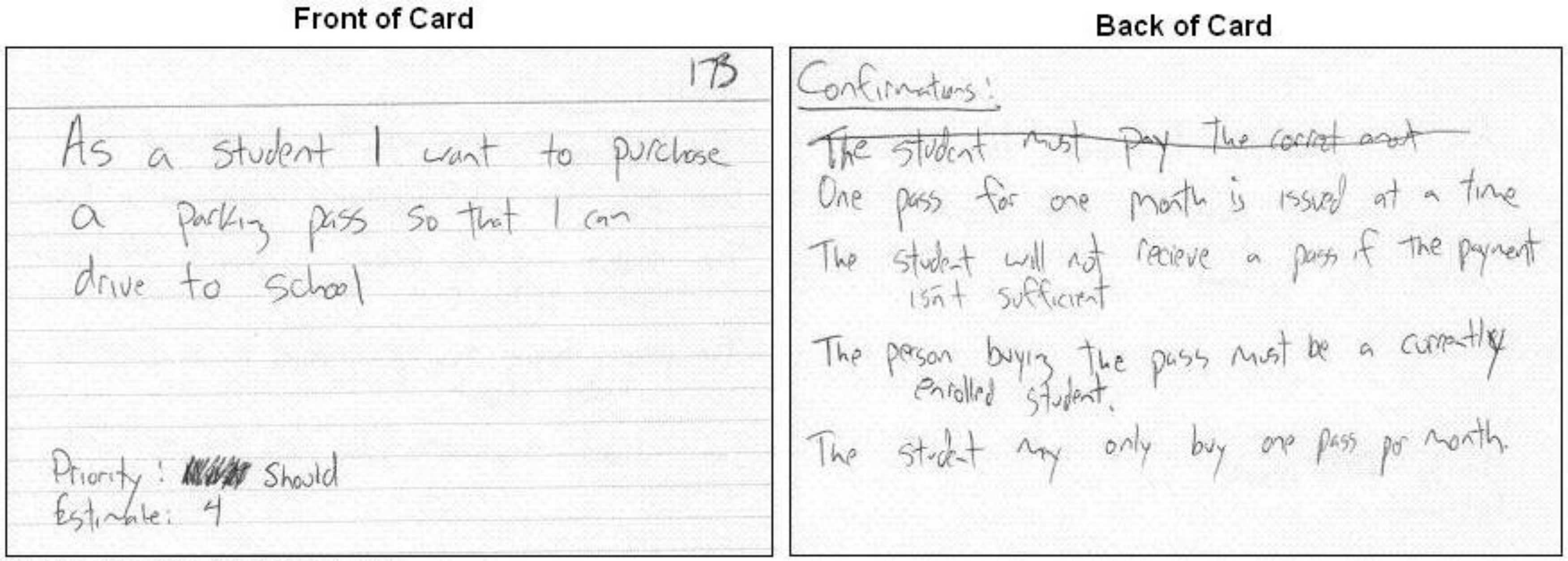
User stories



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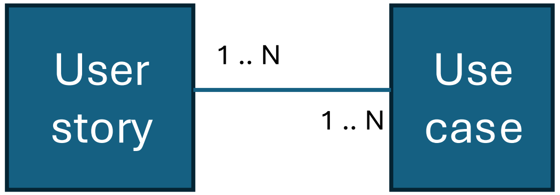
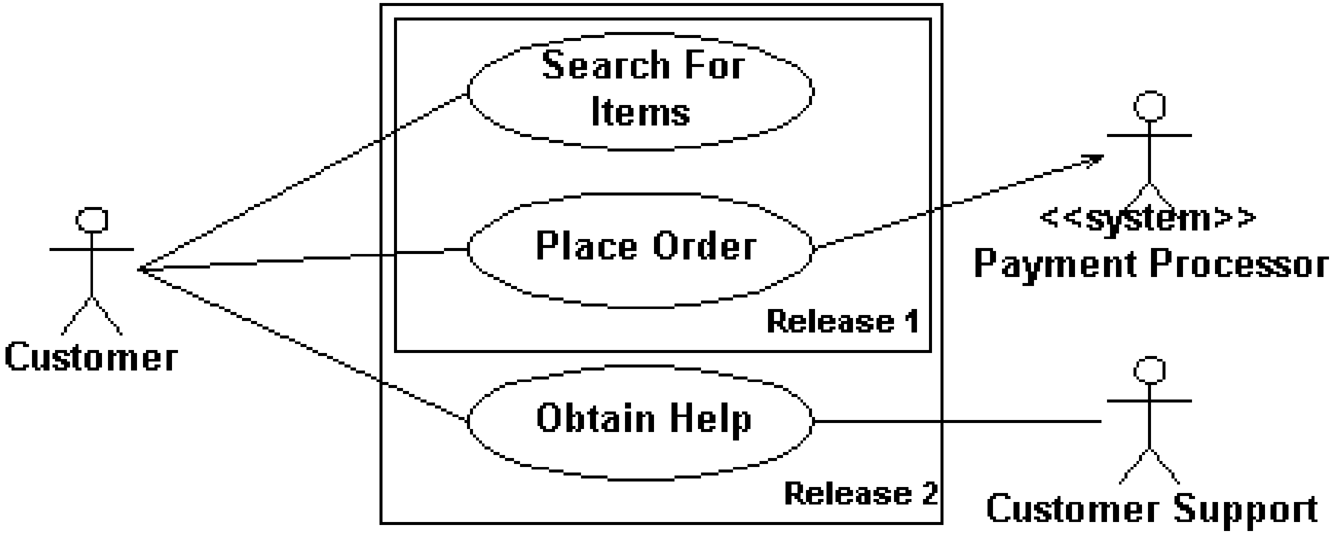
Uddybning af user stories med use cases

Mål	**”Vis etage”.**		
Initiering	Aktiveres af gæst.		
Aktører (Primær, sekundær)	Ekstern bruger, Medarbejder		
Frekvens	500-1.000 visninger pr. uge	Antal samtidige forekomster	-
Ikke-funktion krav	Skal kunne vises på under 1 sekund		
Referencer/Bemærkninger			
Startbetingelser	-		
Beskrivelse	1. Aktør vælger ”Vis etage” 2. System præsenterer hvilken etage aktøren vil begive sig til.		
Variationer	-		
Undtagelser	-		
Slutresultat	Succes	Aktør får af system oplyst, på hvilken etage elevatoren er på vej til	
	Fejl	Aktør kan ikke få oplyst af system, hvilken etage elevatoren begiver sig til	



Akrører og use case

Use case diagram



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Kvalitetssikring

Måling af kvalitet
Funktionelle krav

- Acceptkriterier (bagsiden af user story)
- Use cases
- Nedbrydning/konkretisering af user stories
- Test cases

Typer af non-funktionelle krav

Rettet mod

- aktører
- software produktet

Product maintenance

- Maintainability (can I fix it?)
- Flexibility (can I change it?)
- Testability (can I test it?)

Product operation

- Correctness (does it do what I want it to do?)
- Reliability (does it do it accurately all the time?)
- Efficiency (will it run on my hardware as well as it can?)
- Integrity (is it secure?)
- Usability (can I operate it?)

Product transition

- Portability (will I be able to use it on another machine?)
- Reusability (will I be able to reuse some of the software?)
- Interoperability (will I be able to interface it will some other system?)

Måling af kvalitet
Non-funktionelle krav

Brugervenlighedskrav	**Effektivitets/performance krav**
Systemet skal være let at bruge	Kunne præsentere et søgeresultat på 10 s.
4 ud af 5 personer skal kunne finde en koncert og købe en billet på under fem minutter	Kunne håndtere 100 søgninger/minut (spidsbelastning), 30 søgninger/minut (snit)
En undersøgelse blandt xxx tilfældigt udvalgte brugere skal resultere i at mindst 80% finder systemet nyttigt og let at bruge	Håndtere 100 aktive brugere
80% af testpersonerne skal kunne løse opgave zz på 2 minutter uden alvorlige fejl. To uger senere skal 70% af personerne kunne løse samme opgave uden fejl.	Vise planlagte begivenheder 6 måneder frem i tiden
	Gemme data om begivenheder 12 måneder bagud i tiden

Soren Lauesen: Software Requirements © Addison-Wesley 2002

Entydige? Meningsfulde? Hvor kommer tallene fra? Kan kravene opfyldes? Kan de verificeres? Hvornår?

Vandfaldsmodellen

Kvalitetssikring ved faseovergange

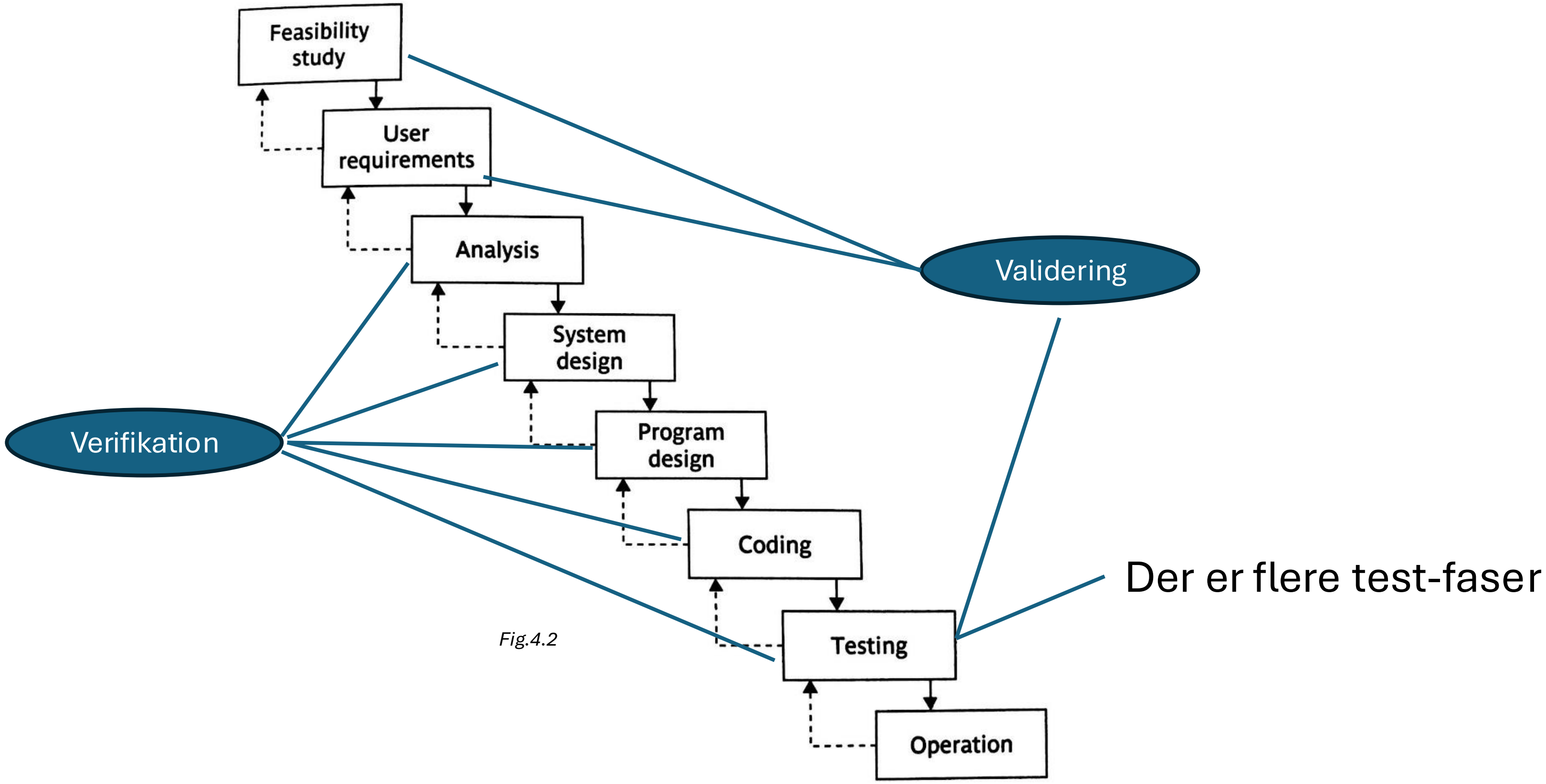
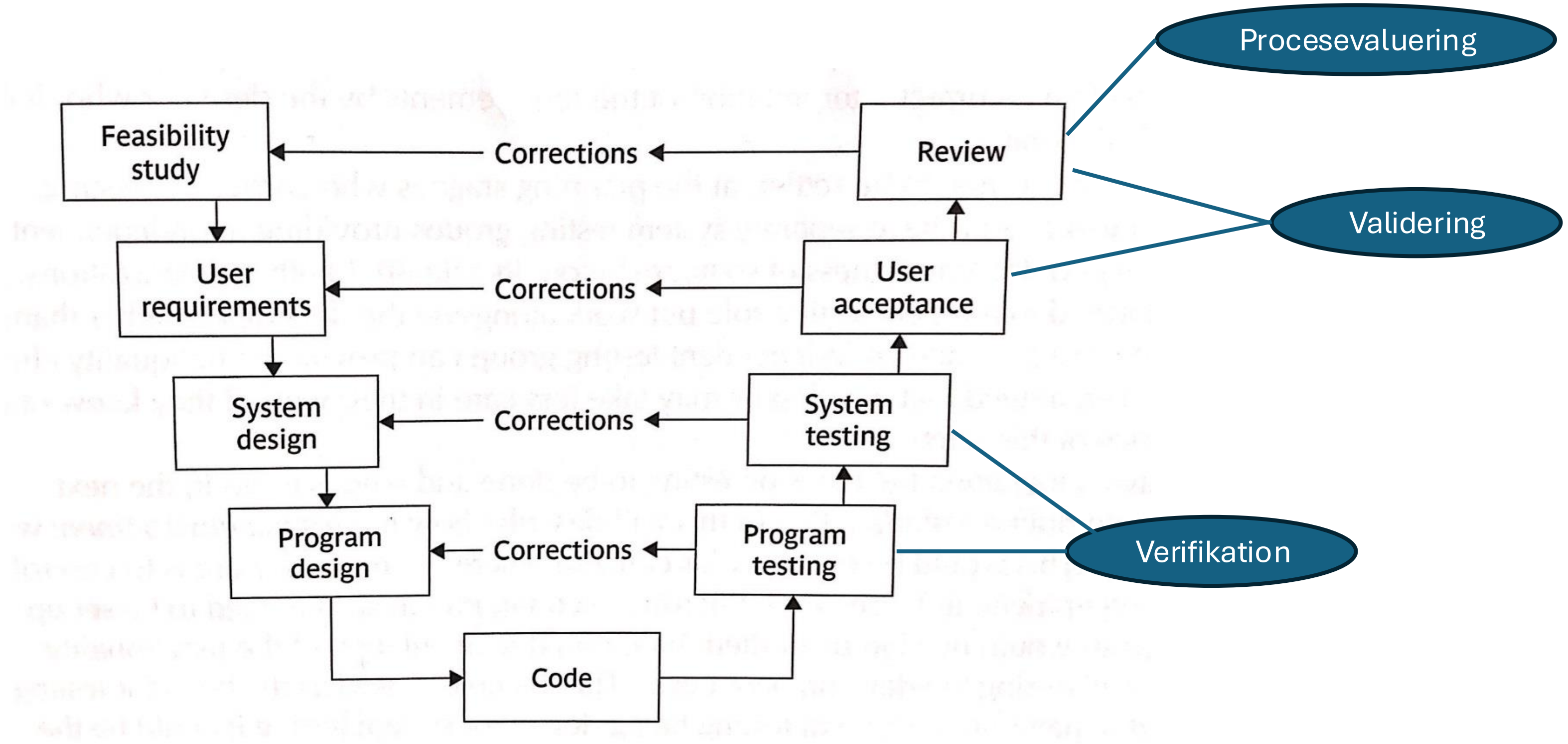


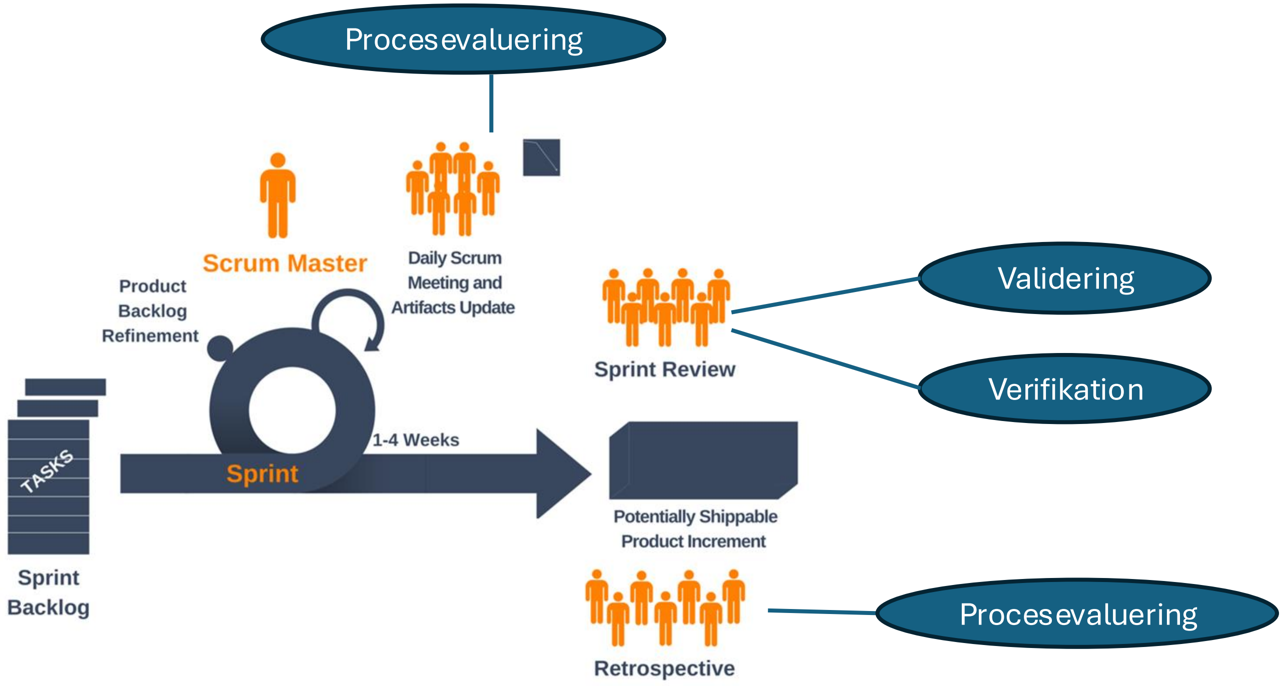
Fig.4.2

V-modellen

Sammenhængen mellem testfaser og analyse/design



Scrum
Kvalitetssikring under og efter Sprint



SCRUM

Kilde: yanado.com

Sprint Reviews

Project: I2C2 pharmacy	**Team:** Pharmacy team	**Sprint:** 1
Stories planned	**Demo feedback**	**Next sprint actions**
COMPLETE		
Story 1	Story 1: meets needs	
Story 3	Story 3: needs clearer error messages	Story 3: Add 1 day of work to Sprint 2
Story 4	Story 4: requested interim save feature (nice-to-have)	Story 4: Move to testing, add new feature to nice-to-haves
Story 5	Story 5: meets needs	
Story 7	Story 7: need to add three data elements	Story 7: Add 2 days of work to Sprint 2
PARTIAL		
Story 2 (demoed)	Story 2: partial demo design is fine	Story 2: Finish remaining task in Sprint 2
Story 9 (no demo)	Story 9: realized logic more complex than first understood	Story 9: Defer past Sprint 2 while logic is clarified
NOT DELIVERED		
Story 6	Story 6: Ran out of time in sprint	Story 6: Move to Sprint 2
Story 8	Story 8: Ran out of time in sprint	Story 8: Story this "nice-to-have" story in the backlog

First of two end-of-sprint "inspect-and-adapt" opportunities

- Focuses on demo of working software to Product Owner and other business stakeholders
- Key questions
  - What did we plan to deliver?
  - What did we actually deliver?
  - How well do delivered stories meet business need?
  - What should we plan for next sprint?

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Sprint Retrospectives

Project: I2C2 Pharmacy	**Team:** Pharmacy Team	**Sprint:** 1
+ PLUS **What Went Well**	**Δ DELTA** **What We Can Improve**	🎬 **ACTIONS** **To Take in Next Sprint**
1. Effective requirement gathering 2. Good client communication 3. Effective planning poker estimating 4. Kept good work efficiency throughout the sprint 5. Did not add “extra” work from client once sprint was launched 6. Finishing one story before starting the next 7. Client satisfaction after each sprint	8. Overallocated the workload in the sprint 9. Requirements in specific stories missed key alternative scenarios 10. Two team members missed standups because of outside meetings	8. Allocate fewer days of work in Sprint 2 9. Revisit these stories with client, with additional supporting BA involvement to ensure 10. Scrum master work to reschedule the outside meetings

Second of two end-of-sprint “inspect-and-adapt” opportunities

- Focuses within team on effectiveness of software construction process
- Key questions:
  - \+ PLUS what went well?
  - Δ DELTA what can we improve?
  - 🎬 ACTIONS to take in next sprint?

Teknikker

- Statiske
  - Sammenligning af dokumenter (krav <-> design <-> program)
    - Inspektion
    - Formelle specifikationer
  - Vurdering af dokumenter (inkl. programmet)
    - Inspektion
    - Walkthrough
    - Metrikker til kodeevaluering
- Dynamiske
  - Modultest
  - Integrationstest
  - Belastningstest
  - ....

Planlægning af kvalitetssikring

- Konsekvenser for projektplanen
  - Forberede, afholde og dokumentere inspections og tests
    - Hvornår? Hvem
  - Vurdere og evt. rette fejl, kvalitetssikre rettelsen mv.
    - I dokumenter og kode
    - I processen

Klynge 2

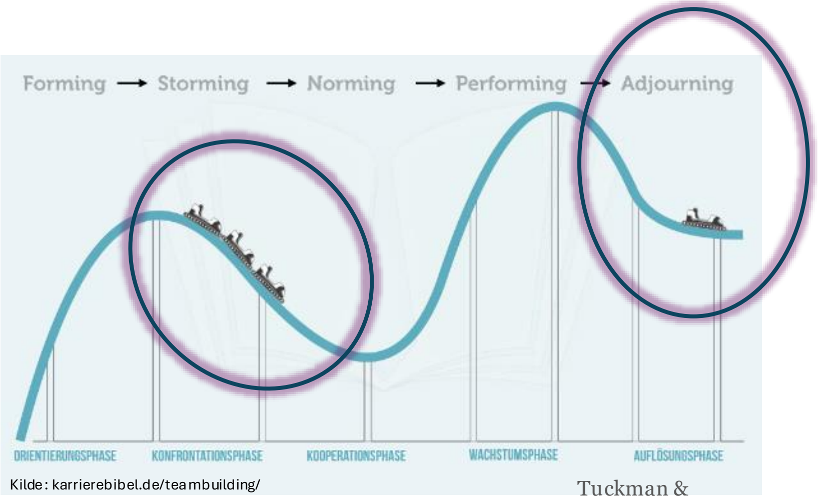
Aflevering til klynge 2

- Kravspecifikation
  - I det format I vælger
- Arkitektur
  - Broen mellem ITP-rapporten og afleveringen i Programmering
  - Beskrivelse af designet og arkitekturen for jeres løsning
  - I kan/skal bl.a bruge det til at tage afsæt i formuleringen af de non-funktionelle krav

Procesevaluering

Den endelige rapport

Proces evaluering



Tuckman & Jensen

5 typiske faser i teamet

Adjourning:
Væsentlig for at deltager 'kan komme videre'

Refleksion Belbins roller ift gruppedynamikken

Proces evaluering

Emner til evalueringsmøde Dato: Udfyldt af:		
Det faglige samarbejde interne interessenter I projektgruppen		
Det faglige samarbejde eksterne interessenter Udenfor projektgruppen - i og udenfor virksomheden		
Styring i Projektet Hvordan har projektlederen generelt styret projektet Tidsplanen (estimat/opfølgning) Kommunikation Organisering		
Produktet Leverancer iht. overdragelse		
Anvendelse af div. modeller /værktøjer Herunder projektmodel		

Nævn min. 3 af de største udfordringer du har haft i projektet	
Nævn min. 3 af de største positive oplevelser du har haft i projektet	

Større projekter af en vis varighed

Evaluering i team … og i styregruppe

Evt. ved projektgruppemøde

Forberedelse

Dialog og opsamling af læring

Midtvejsevaluering

Hvad fungerer godt?

Hvad skal vi gøre anderledes/tilføje?

Hvad skal vi holde op med?